

# Chapter 1

- Multinational Corporations and Financial Management

# Chapter Outline

- We live in a global world
- MNCs and internationalization
- Consequences of global competition
- The value of trade

# We live in a global world

- Globalization and business
- What is an MNC?
- Multinational products
- MNCs by country
- American MNCs

# Globalization and Business (1 of 2)

- **The global marketplace is not for amateurs.**
  - Fast-paced and unpredictable – e.g., Nokia, once the world leader in cell phones, was displaced by Apple and Samsung in just a few years.
  - Requires specialized knowledge and understanding – e.g., how do companies manage interest rate exposures?
  - You probably will, or already do, work for a firm with international exposure or a non-US multinational company (“MNC”).

## HQ in France with large US presence

| MNC            | Business        |
|----------------|-----------------|
| Michelin       | Tires           |
| Sanofi         | Pharmaceuticals |
| Saint-Gobain   | Glass           |
| Bic            | Pens            |
| Accor          | Hotels          |
| Renault-Nissan | Automobiles     |

## HQ in Germany with large US presence

| MNC           | Business        |
|---------------|-----------------|
| Siemens       | Industrial mfg. |
| BMW           | Automobiles     |
| Mercedes Benz | Automobiles     |
| SAP           | Software        |
| Bayer         | Pharmaceuticals |

# Where are these products from?



# Multinational Products (1 of 3)

## Barbie

|                                    |                                                                                   |
|------------------------------------|-----------------------------------------------------------------------------------|
| <b>Designed in:</b>                | US (California)                                                                   |
| <b>Parts and clothing origins:</b> | Japan, China, Hong Kong, Malaysia, Indonesia, Korea, Italy, and Taiwan            |
| <b>Assembly:</b>                   | Mexico                                                                            |
| <b>Customers:</b>                  | 60% North America<br>25% Europe<br>11% Latin America<br>6% Asia Pacific<br>(2022) |



# Multinational Products (2 of 3)

## Apple iPhone

|                             |                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Designed in:</b>         | US (California)                                                                                                                                                                    |
| <b>Parts and suppliers:</b> | Australia, Brazil, China, Denmark, France, Germany, India, Israel, Italy, Japan, The Netherlands, South Korea, Malaysia, Mexico, South Africa, Spain, Taiwan, Turkey, the UK, etc. |
| <b>Assembly:</b>            | China and India                                                                                                                                                                    |
| <b>Customers:</b>           | 130 countries                                                                                                                                                                      |



# Hollywood relies on a global market

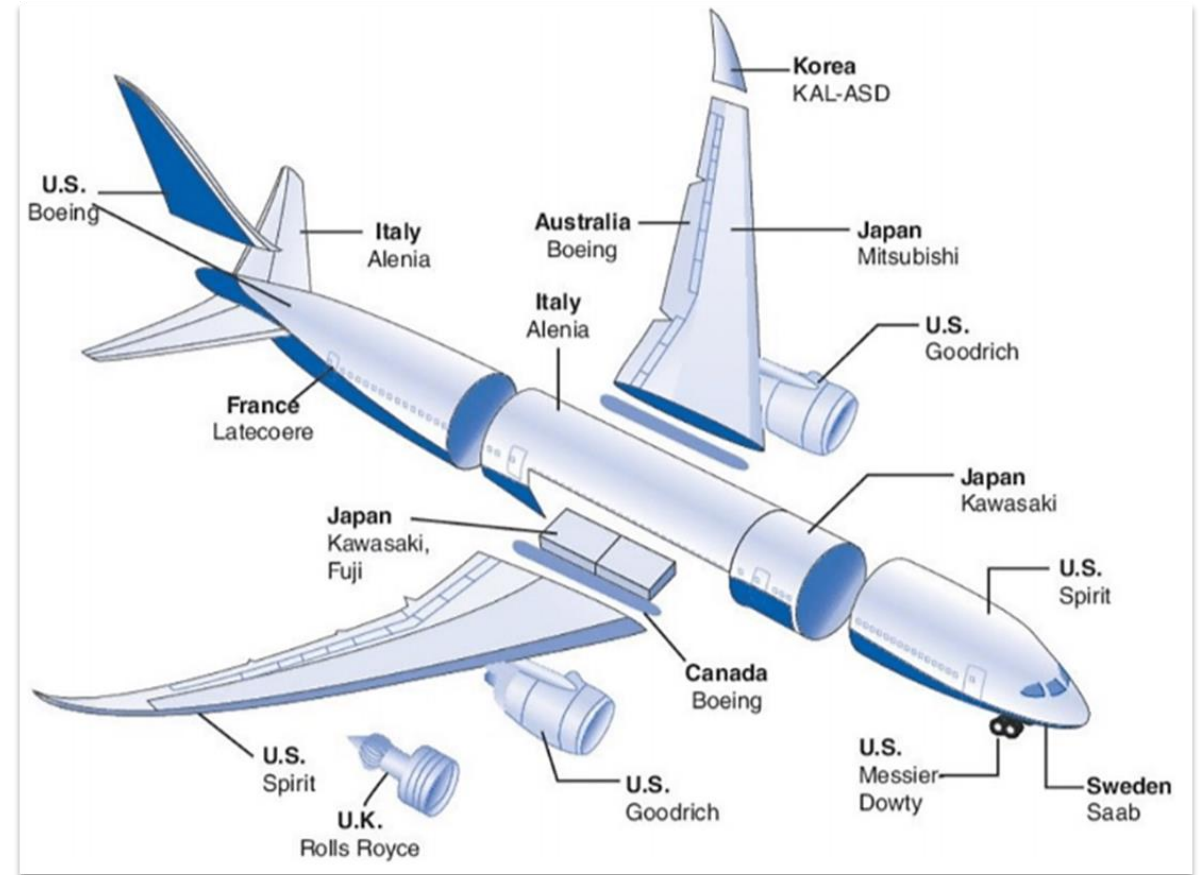
|                                                                                     |                                                                     |                                                                                                |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
|    | Producer: Michael Wilson                                            | British-American                                                                               |
|    | Director: Sam Mendes                                                | British                                                                                        |
|    | Cinematography: Hoyte van Hoytema                                   | Dutch-Swedish                                                                                  |
|    | Actors: Daniel Craig, Christoph Waltz, Lea Seydoux, Monica Bellucci | British, Austrian-German, French, Italian                                                      |
|    | Location                                                            | United Kingdom, Italy, Morocco, and Mexico                                                     |
|  | Distribution                                                        | Columbia Pictures (USA) subsidiary of Sony (Japan)                                             |
|  | Sales                                                               | \$880.7 million worldwide; \$135.5 million from the UK; and \$200.1 million from North America |





# Multinational Products

- Sources of parts for Boeing's 787 Dreamliner

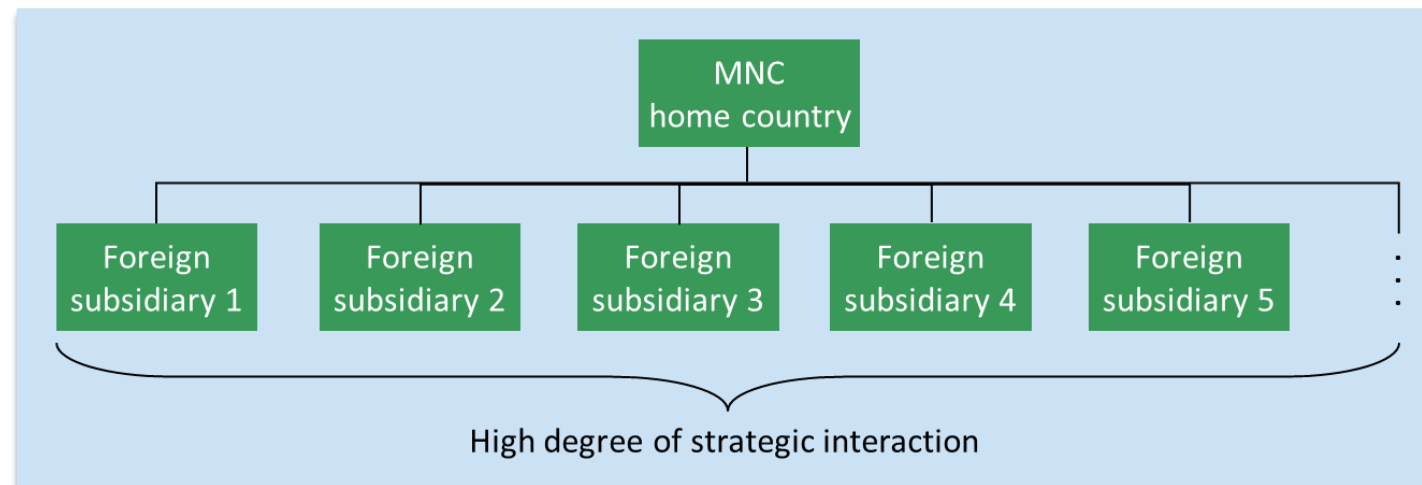


# History of the Multinational Companies

- 16<sup>th</sup> to 18<sup>th</sup> century
  - British East India Company
  - The Dutch East India Company
  - The Hudson Bay Company
- 1868-1870 beginning of the modern MNCS

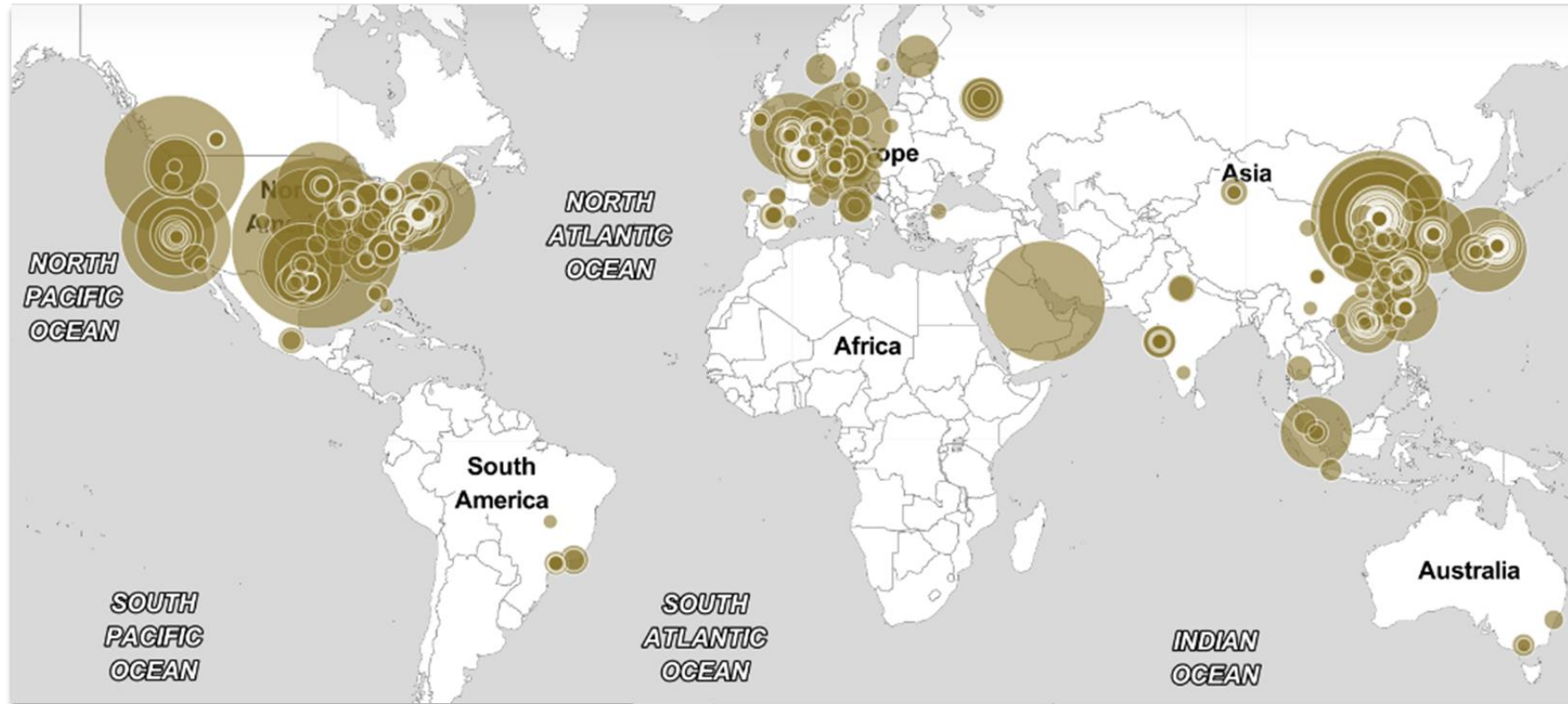
# What is an MNC?

- MNCs produce and sell goods or services in more than one country.
- MNCs emphasize group performance over performance of individual parts.



- What differentiates MNCs from other firms engaged in international business is the ***globally coordinated allocation of resources by a single centralized management.***

# MNCs By Country (1 of 2)



Fortune Global 500 MNCs (2022) are heavily concentrated in North America, Europe and East Asia.

# MNCs by Country (2 of 2)

**Fortune Global 500 companies by top 10 counties, August 2022**

| Rank | Country                                                                             |                | Companies |
|------|-------------------------------------------------------------------------------------|----------------|-----------|
| 1    |    | China          | 145       |
| 2    |    | United States  | 122       |
| 3    |    | Japan          | 47        |
| 4    |    | Germany        | 28        |
| 5    |    | France         | 25        |
| 6    |    | United Kingdom | 23        |
| 7    |   | South Korea    | 16        |
| 9    |  | Switzerland    | 14        |
| 9    |  | Canada         | 14        |
| 10   |  | India          | 13        |

# American MNCs

- American MNCs comprise <1% of US firms.
  - But still account for
  - 23% of total private industry employment,
  - 21% of GDP,
  - >50% of exports, and
  - 70% of R&D spending.
- As a group, derive a high proportion of their revenues abroad.

| Company/Brand                                                    | % foreign revenues |
|------------------------------------------------------------------|--------------------|
| Alphabet<br>Apple<br>Facebook<br>Boeing<br>Intel<br>Pfizer       | >40%               |
| Coca Cola<br>Johnson & Johnson                                   | ~65%               |
| Mondelez (Oreo,<br>Trident gum,<br>Philadelphia cream<br>cheese) | >74%               |

# Top 20 MNCs by Market Capitalisation

Multinational corporation (MNC) rankings are determined by distinct metrics: **market capitalisation** (value of outstanding shares) or **revenue**.

The world's **top 20 largest MNCs by market capitalisation** are:

**Alphabet** (United States)

**Apple** (United States)

**Microsoft** (United States)

**Nvidia** (United States)

**Amazon** (United States)

**Meta Platforms** (United States)

**Saudi Aramco** (Saudi Arabia)

**Broadcom** (United States)

**Tesla** (United States)

**TSMC** (Taiwan)

# Top 11 to 20 MNCs

The **top 20 MNCs ranked by market capitalisation** globally include the above, plus

- 11. **Berkshire Hathaway** (United States)
- 12. **Eli Lilly** (United States)
- 13. **JPMorgan Chase** (United States)
- 14. **Walmart** (United States)
- 15. **Novo Nordisk** (Denmark)
- 16. **Tencent** (China)
- 17. **Mastercard** (United States)
- 18. **Exxon Mobil** (United States)
- 19. **Johnson & Johnson** (United States)
- 20. **Netflix** (United States)



# Selected Large U.S. Multinationals

| US Corporations in the World's Top 100 Multinational Companies, Ranked by Foreign Assets, 2019 |                              |                                           |                       |                     |                       |                     |            |                     |
|------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------|-----------------------|---------------------|-----------------------|---------------------|------------|---------------------|
| Rank,<br>Foreign<br>Assets                                                                     | Corporation                  | Industry                                  | Assets                |                     | Sales                 |                     | Employment |                     |
|                                                                                                |                              |                                           | Foreign<br>(billions) | Percent of<br>Total | Foreign<br>(billions) | Percent of<br>Total | Foreign    | Percent of<br>Total |
| 11                                                                                             | Chevron Corp                 | Petroleum Refining and Related Industries | \$172.8               | 72.8%               | \$75.6                | 53.9%               | 22,800     | 47.3%               |
| 13                                                                                             | Exxon Mobil Corporation      | Petroleum Refining and Related Industries | \$169.7               | 46.8%               | \$123.8               | 48.4%               | 35,058     | 46.8%               |
| 22                                                                                             | Microsoft Corporation        | Computer and Data Processing              | \$117.5               | 41.0%               | \$61.6                | 49.0%               | 59,000     | 41.0%               |
| 23                                                                                             | Johnson & Johnson            | Pharmaceuticals                           | \$116.2               | 73.7%               | \$39.2                | 47.7%               | 97,393     | 73.7%               |
| 24                                                                                             | Apple Computer Inc           | Computer Equipment                        | \$114.7               | 33.9%               | \$157.9               | 60.7%               | 47,928     | 35.0%               |
| 25                                                                                             | General Electric Co          | Industrial and Commercial Machinery       | \$112.7               | 42.4%               | \$55.9                | 58.7%               | 135,000    | 65.9%               |
| 45                                                                                             | IBM Corp                     | Computer and Data Processing              | \$82.4                | 54.1%               | \$40.8                | 53.0%               | 190,885    | 54.1%               |
| 50                                                                                             | Pfizer Inc                   | Pharmaceuticals                           | \$76.3                | 45.5%               | \$27.9                | 53.9%               | 40,211     | 45.5%               |
| 52                                                                                             | Ford Motor Company           | Motor Vehicles                            | \$75.8                | 29.3%               | \$57.1                | 36.7%               | 90,000     | 47.4%               |
| 53                                                                                             | Walmart inc                  | Retail Trade                              | \$74.6                | 31.6%               | \$121.4               | 23.2%               | 700,000    | 31.8%               |
| 59                                                                                             | Amazon.com, Inc              | E-Commerce                                | \$67.2                | 29.8%               | \$85.4                | 30.4%               | 237,953    | 29.8%               |
| 64                                                                                             | Alphabet Inc                 | Computer and Data Processing              | \$63.6                | 23.0%               | \$87.0                | 53.8%               | 27,403     | 23.0%               |
| 65                                                                                             | Procter & Gamble Co          | Chemicals and Allied Products             | \$61.1                | 53.1%               | \$39.1                | 57.8%               | 51,460     | 53.1%               |
| 75                                                                                             | The Coca-Cola Company        | Food & beverages                          | \$54.1                | 62.5%               | \$25.5                | 68.6%               | 76,100     | 88.3%               |
| 77                                                                                             | Raytheon Technologies Corp   | Aircraft                                  | \$53.6                | 38.3%               | \$29.7                | 38.6%               | 93,215     | 38.3%               |
| 78                                                                                             | Mondelez International, Inc. | Food & beverages                          | \$53.1                | 82.3%               | \$19.2                | 74.4%               | 68,000     | 85.0%               |
| 81                                                                                             | Intel Corporation            | Electronic components                     | \$49.6                | 36.3%               | \$56.3                | 78.3%               | 56,508     | 51.0%               |
| 97                                                                                             | General Motors Co            | Motor Vehicles                            | \$45.0                | 19.7%               | \$26.6                | 19.4%               | 68,000     | 41.5%               |

Source: United Nations Conference on Trade and Development

# MNCs and Internationalization

- Why corporations internationalize
- Measuring internationalization
- How firms become MNCs
- MNC risk exposure

# Why Corporations Internationalize (1 of 2)

## **1. Find raw materials**

Not all raw materials exist domestically (oil, metals, minerals, etc.).

## **2. Find new markets**

Main motive of modern MNC (McDonald's, P&G, Coca-Cola).

## **3. Reduce costs**

Exploit lower overhead, taxes, labor, and environmental costs and lower levels of regulation.

# Why Corporations Internationalize (2 of 2)

## Find raw materials

Early 1600s: The first “corporations” began as trading companies in search of exotic goods: The British East India Company, The Dutch East India Company, The Hudson Bay Company, ...

## Find new markets

1853: Colt Corporation (firearms) starts a London, UK, factory in 1853

1867: Singer Corporation (sewing machines) starts a Glasgow, UK, factory

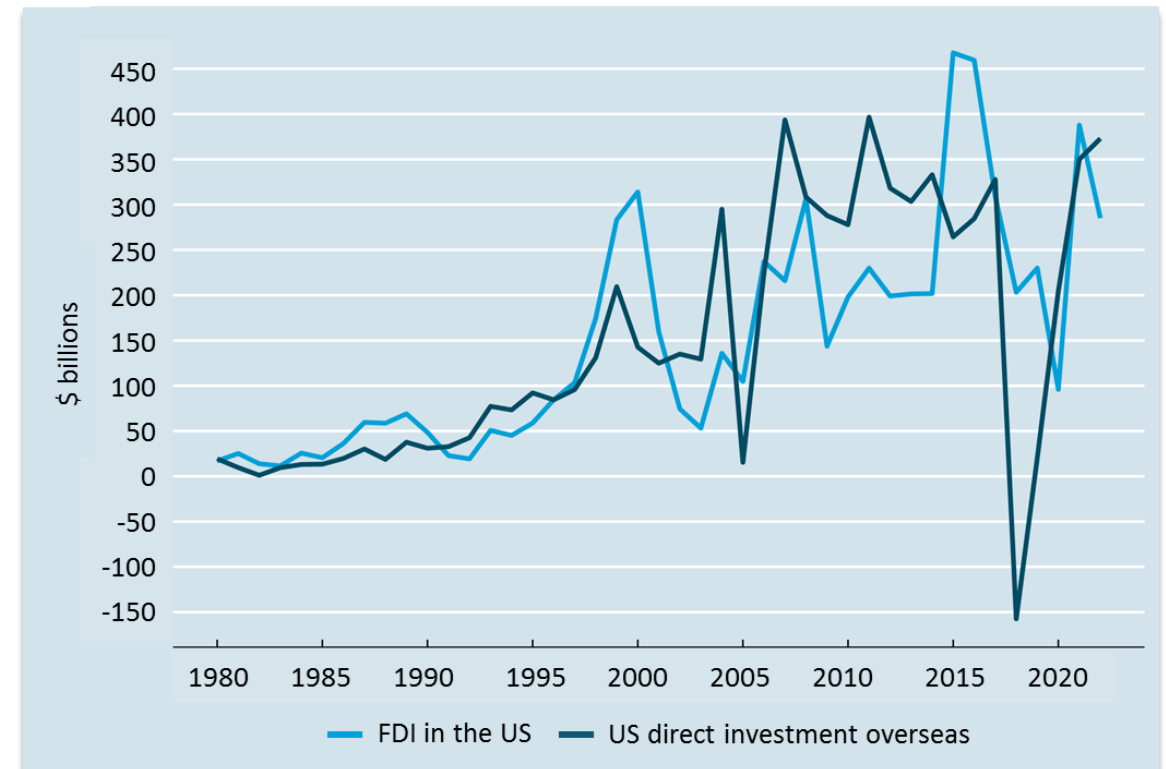
## Reduce costs

1960s: Many MNCs started manufacturing in Mexico to reduce labor costs (incl. Asahi Overseas Corp., Bayer Corp., Eli Lilly & Co., Emerson Electric Co., General Dynamics Corp., and Hughes Corp.)

# Measuring Internationalization (1 of 3)

- **Why MNCs have expanded since 1970s/1980s**
- Foreign direct investment (“FDI”) is a measure of growing internationalization.
- Overseas investment by US firms (FDI outflows) and US investment by foreign firms (FDI inflows) are in the hundreds of billions of dollars each year.
- In 2018, American businesses made large repatriations of earnings from their foreign affiliates following the US Tax Cuts and Jobs Act (2017), causing US investment overseas to plummet to -\$157 billion.

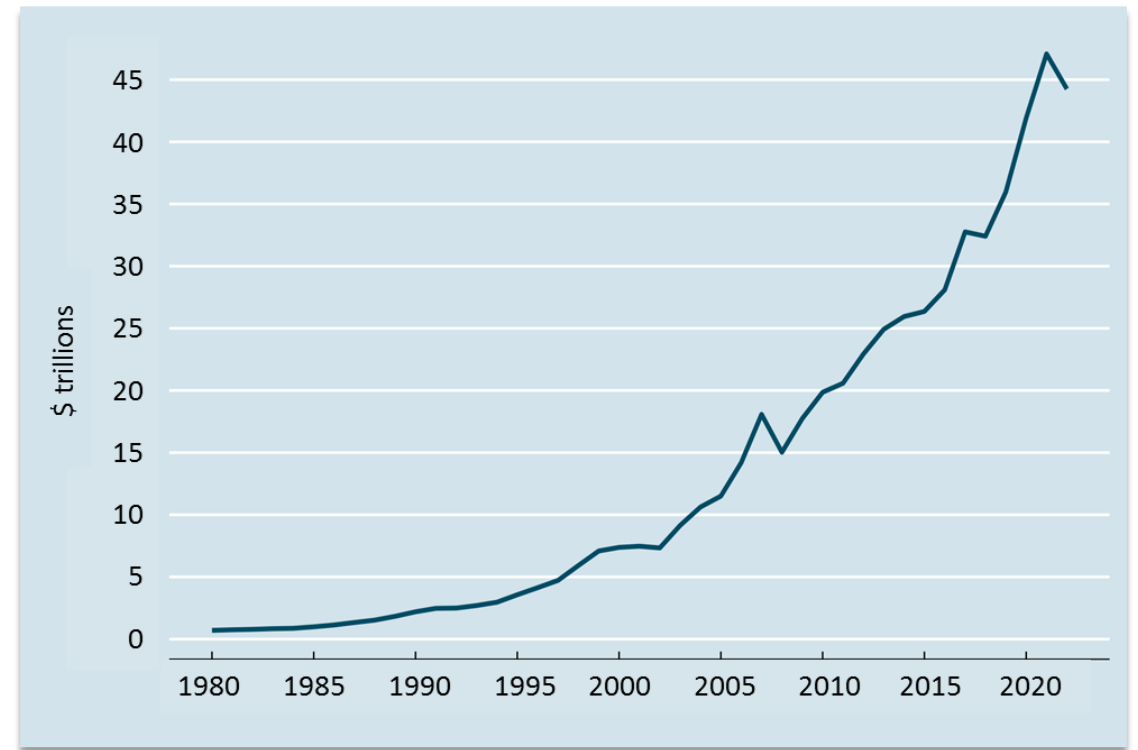
**Annual US foreign direct investment inflows and outflows, 1980-2022**



## 3.2 Measuring Internationalization (2 of 3)

- The stock of US FDI outflows reached \$6.6 trillion in 2022 while the stock of US FDI inflows on a comparable basis exceeded \$5.2 trillion.
- Worldwide, the stock of FDI was \$44.25 trillion in 2022.
- The long-term trend is upwards, due to:
  - Falling regulatory barriers to overseas investment;
  - Rapidly declining telecommunications and transport costs; and
  - Freer domestic and international capital markets in which vast sums of money can be raised, companies can be bought, and currency and other risks can be hedged.

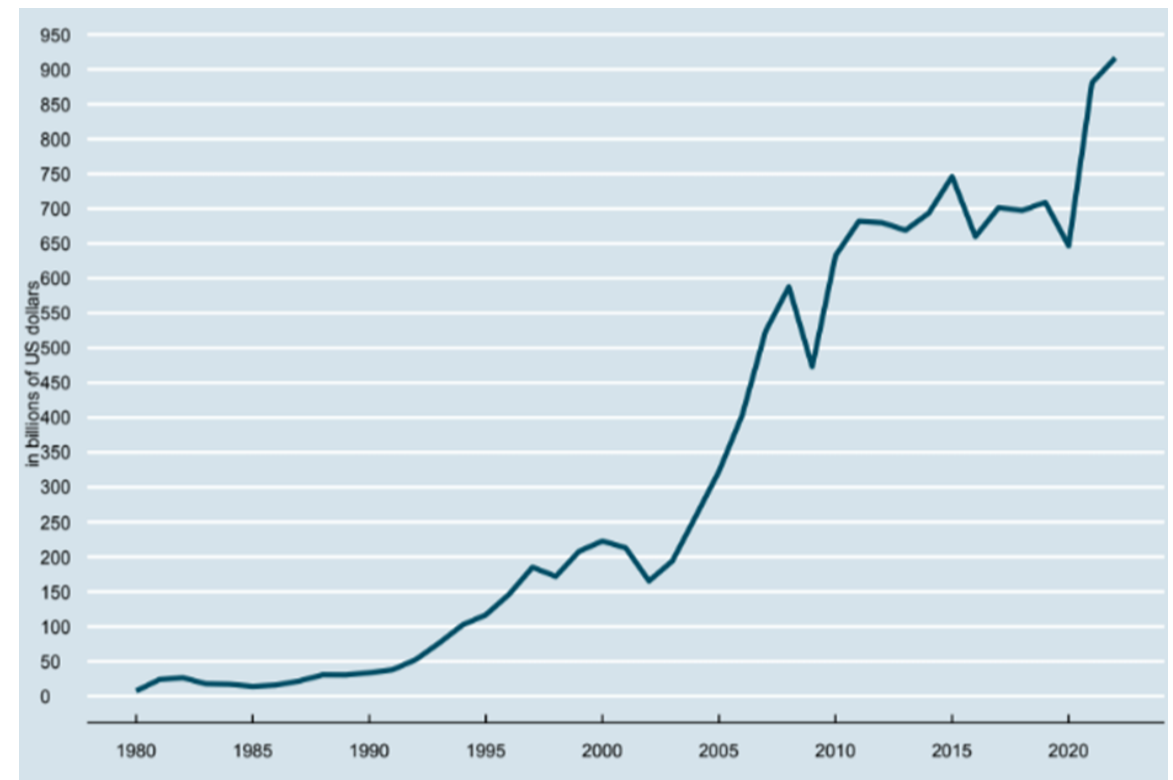
**Stock of worldwide FDI, 1980-2022**



# Measuring Internationalization (3 of 3)

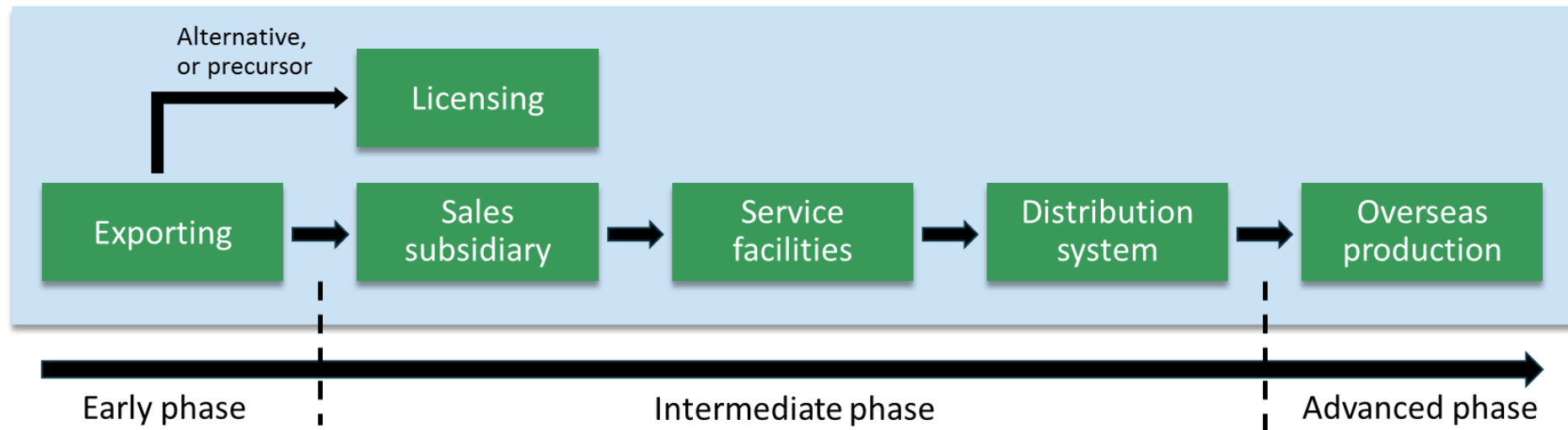
- **Expansion into emerging markets** enables firms to achieve economies of scale and exploit the premium associated with their brand names.
- **Reduced trade barriers** and regulations by more free-market-oriented governments have reduced the costs and risks of FDI opportunities.
- In response, FDI in emerging markets by MNCs has soared, despite the episodic crises these countries often face.

Annual FDI inflows to developing countries, 1980-2022



## 3.3 How Firms Become MNCs

- Most begin as domestic businesses.
- They become MNCs in *phases* over time.
  - Begin with ***simple, low-risk, low-return*** strategies.
  - Progress to ***sophisticated, high-risk, high-return*** strategies.





# How Firms Become MNCs

## Example: SLB (formerly Schlumberger)

| Timeline | Milestone                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1926     | Founded in France by brothers Conrad and Marcel Schlumberger to map subsurface rock bodies                                                                                                                                                                                                                                                                                                       |
| 1930s    | International expansion into main oil-producing countries; from team of 8 to 140                                                                                                                                                                                                                                                                                                                 |
| 1940s    | First offshore rigs                                                                                                                                                                                                                                                                                                                                                                              |
| 1950s    | New technology, strategic acquisitions                                                                                                                                                                                                                                                                                                                                                           |
| 1960s    | Work with NASA                                                                                                                                                                                                                                                                                                                                                                                   |
| 1980s    | Global research network                                                                                                                                                                                                                                                                                                                                                                          |
| Now      | <p>The world's largest oilfield services company with \$33 billion in revenue. Incorporated in Willemstad, Curacao. Headquarters in Houston and Paris. Current CEO is French, Previous CEO was Norwegian, and Before British.</p> <p>Employs &gt;100,000 people from &gt;140 countries working in &gt;85 countries.</p> <p>Listed on the New York, London, Paris, and Zurich stock exchanges</p> |



# MNC Risk Exposure

## Exchange Rate Risk

- Future exchange rates unknown
- Hence, revenues and costs unknown

## Different Regulations

- Restrictions on capital mobility
- Tax laws

## Political Risk

- Risk that foreign government will *directly interfere with or terminate* MNC operations
- Risk that foreign politics or policy will *affect* operations (labor strikes, conflict, etc.)

How the rest of this Course prepares you to develop the knowledge and skills to be part of the management team of an MNC in the future

Various chapters of this book, in particular from Ch 9 onwards, apply financial risk techniques and hedging instruments that could ensure the operations of MNCs are managed in a way that ensures their resilience, survival, and competitiveness.

# Consequences of Global Competition

- Consequences of Globalization
- Foreign competition
- Perceived job loss
- Perceived wage erosion

# Globalization and Business (2 of 2)

## **Catalysts for globalization**

- Massive deregulation
- The fall of the communist bloc
- Revolution in information technologies
- Dramatic decrease in the transportation costs
- The sale of state-owned firms around the world
- The rise of the market for corporate control and private equity
- Dominant use of English as the Lingua Franca of business

# Debate Over Globalization (1 of 2)

- Criticism
  - Trade does not benefit poor countries.
- Evidence
  - Nations that trade grow richer together, **not** at each other's expense.
  - Common error in “pop-economic analysis”: *Zero-sum game* fallacy

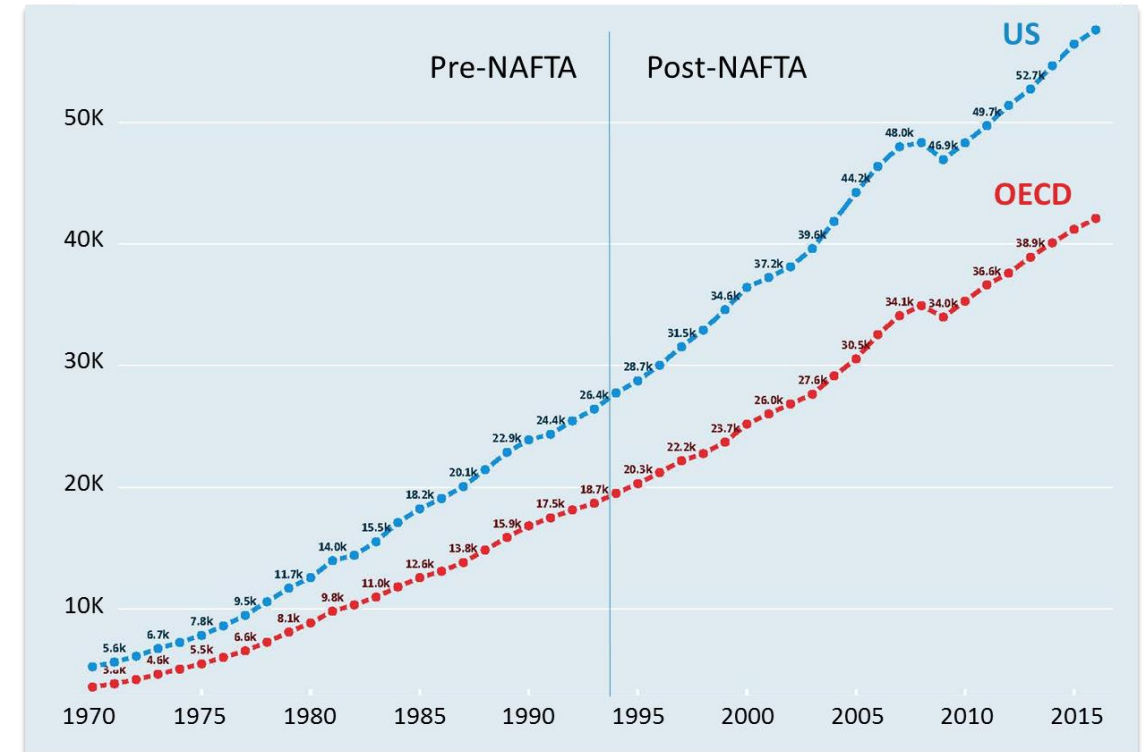
GDP (PPP\*) per person, 1990s growth rates



Source: World Bank

# Debate Over Globalization (2 of 2)

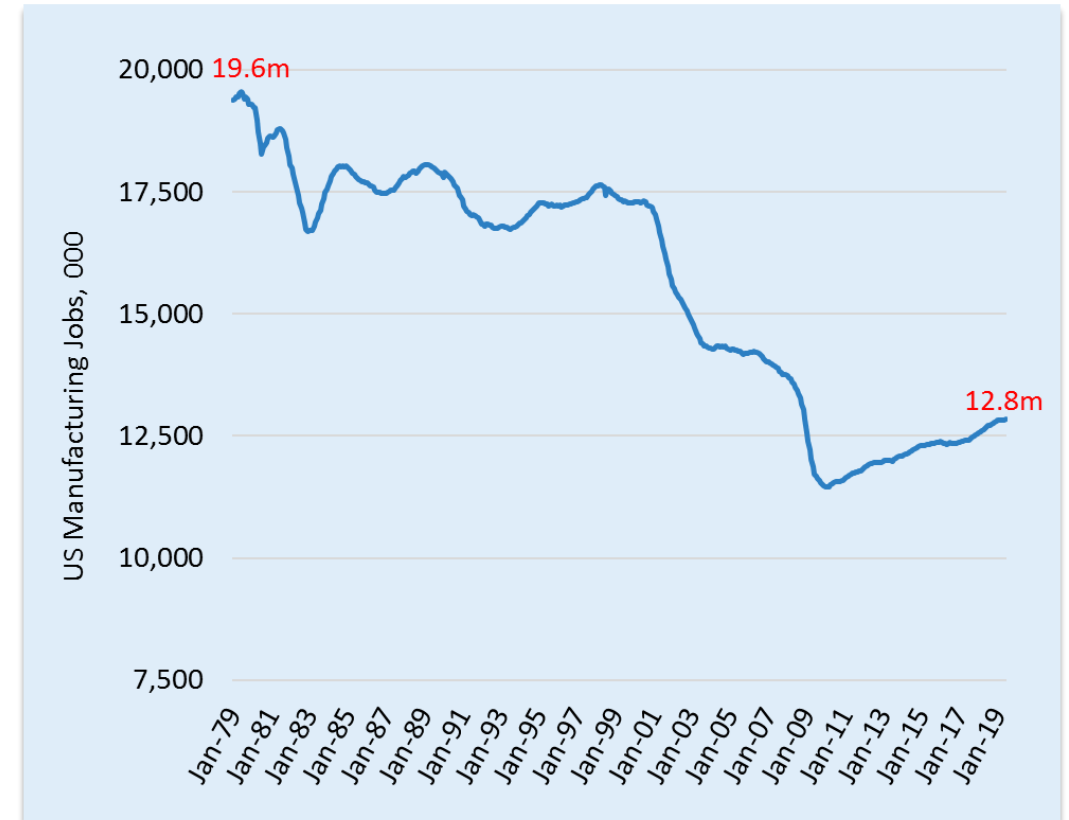
- Criticism
  - Trade does not benefit rich countries countries. NAFTA would cause US jobs to leave to Mexico like a “giant sucking sound”
- Evidence
  - Nations that trade grow richer together, **not** at each other's expense.
  - Common error in “pop-economic analysis”: *Zero-sum game* fallacy



# Debate Over Outsourcing (1 of 4)

## Criticisms

1. Outsourcing is a major source of unemployment.
  - From 1979 to 2019, US manufacturing jobs declined by 34.3%:
    - 19.6 million in 1979
    - 12.8 million in 2019
  - **Assumption:** This decline is due to outsourcing to low-cost nations.
2. Outsourcing yields no benefits to the US.
  - **Assumption:** Outsourcing is a “one-way street,” and US workers pay the ultimate cost.



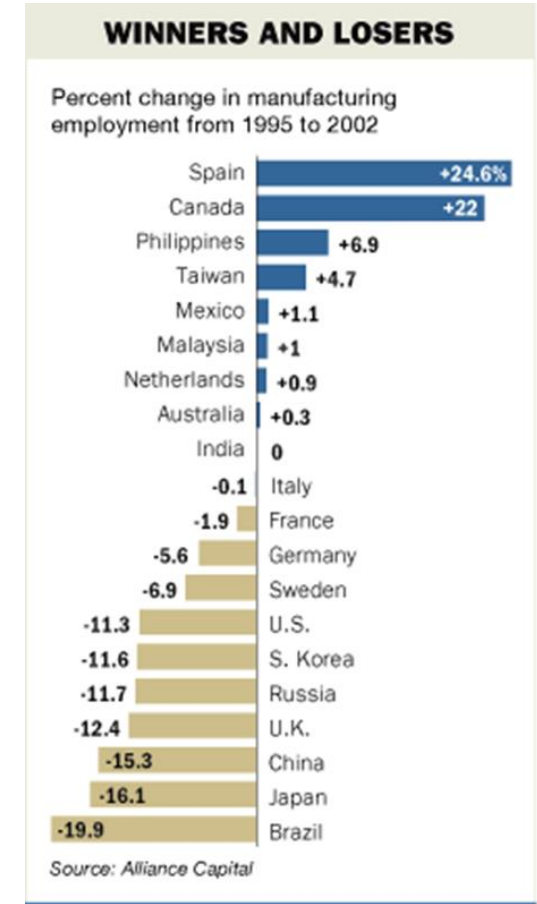
Source: US Bureau of Labor Statistics



# Debate Over Outsourcing (2 of 4)

## Evidence

- Manufacturing jobs have declined worldwide, not just in the U.S.
- In study of 20 economies between 1995 (after NAFTA) to 2002, 22 million factory jobs have been lost.
  - The U.S. lost 11% of its manufacturing jobs.
  - Brazil lost the most jobs (20% of its manufacturing jobs)
  - China lost 15% of its manufacturing jobs (about 15 million jobs)
- The cause of decline in manufacturing jobs: Increased productivity.



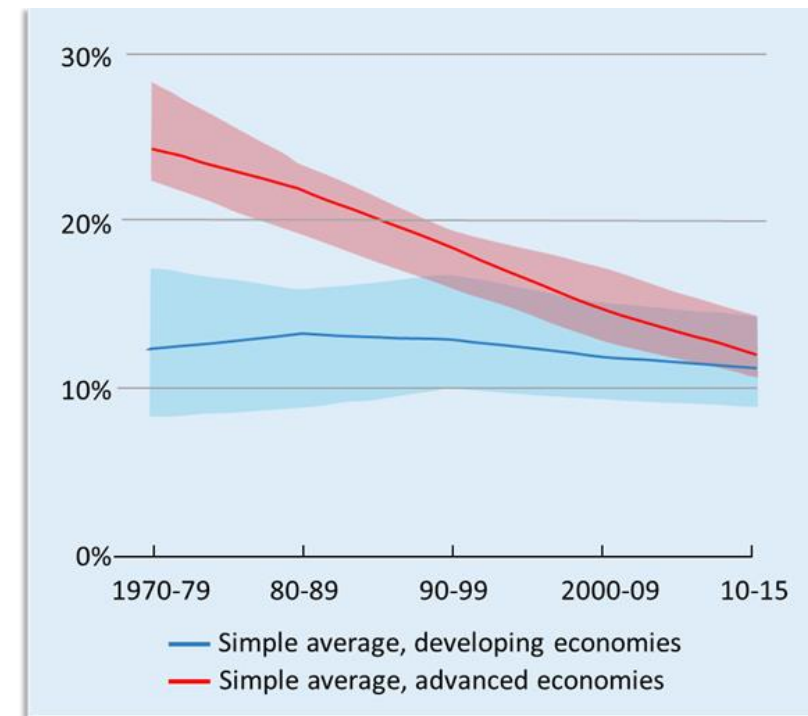
# Debate Over Outsourcing (3 of 4)

## Evidence

- Manufacturing jobs have declined worldwide, not just in the US.
- In an IMF study, the share of global manufacturing jobs has declined in both advanced and emerging economies.
- The real cause of decline in manufacturing jobs is increased productivity.

<https://www.imf.org/wp-content/uploads/2018/04/eng-april-3-weo1-manufacture-2.jpg>

Share of global manufacturing jobs in aggregate employment



Source: International Monetary Fund

Shaded areas are the interquartile range

# Debate Over Outsourcing (4 of 4)

## Evidence

U S “outsourcing” vs. “insourcing”

- In 2020, foreign multinationals employed 8.5 million American workers (2.9 million in the manufacturing sector).

Outsourcing responsible for *<1% of total jobs lost annually*.

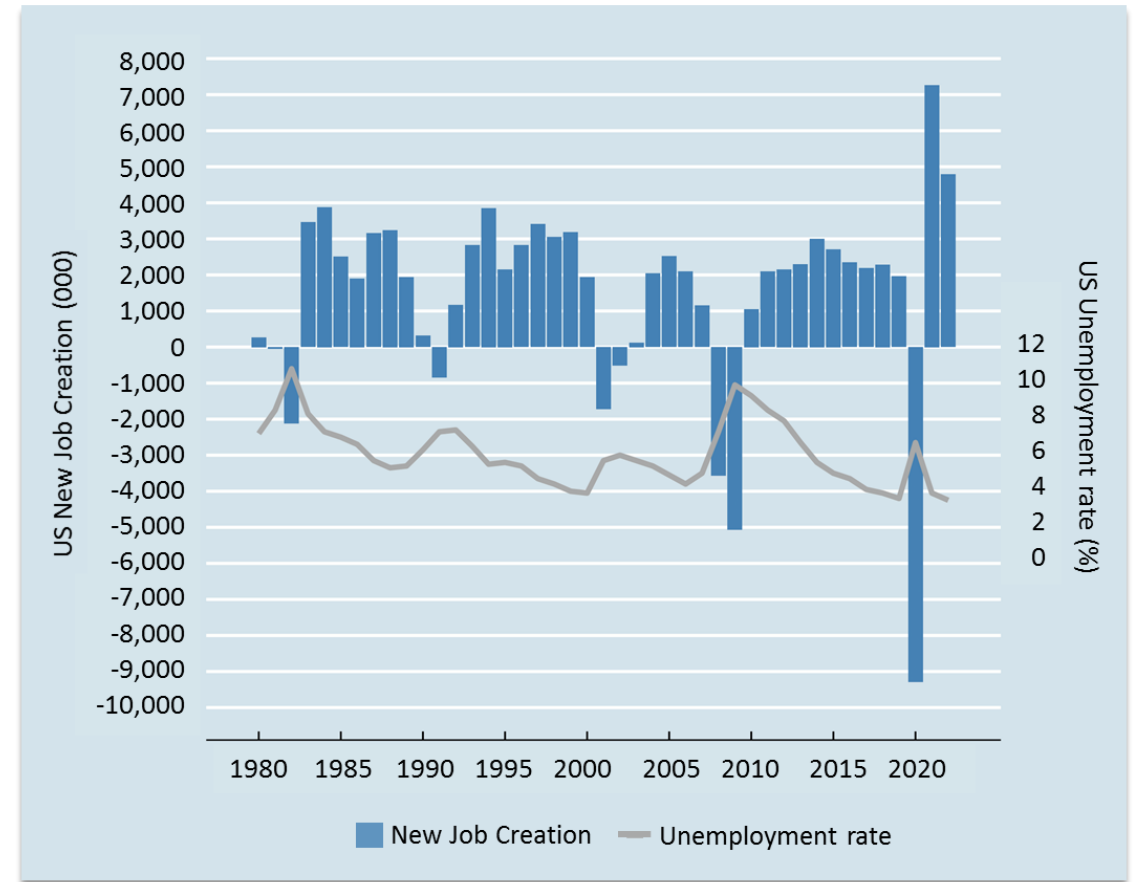
- 100,000 of 15 million jobs lost annually are due to outsourcing.
- On average, 15 million jobs lost replaced by 17 million new jobs.

Law of Comparative Advantage at work!

- Enables higher final output at a lower cost (saves U S firms 30%-50% net in costs).
- Leads to economic growth and higher living standards.

# Perceived Job Loss

- Perceived domestic job losses
  - **Truth:** The U S created approximately 64 million jobs from 1980 to 2022, when competitive pressure from foreign imports was high.
  - Concurrently, unemployment fell during, from 7.2% in December 19 80 to 3.5% in December 2022.



# Perceived Wage Erosion

## Perceived wage erosion

- **Truth:** US real wages have steadily risen.
- Plotting real compensation (wages) against labor productivity (output per hour) from 1960 to 2022, using a base of 100 for 1994—the year NAFTA went into effect.
- As labor productivity has gone up, so have real hourly wages.
- On average and not controlling for other factors that may affect wages, 55% of productivity gains have gone to increase wages.

